Thesis results and revision package

Dictionary-Assisted Hybrid Retrieval-Augmented Generation Assistant for Code-Switched Local Government Services Inquiries in Davao City

Researcher review draft | 28 October 2026

Prepared from archived evaluation records, the 20260923 manuscript, the two-module audit amendment and the Thesis Tracker. Tables and figures report existing results; no model or judge evaluation was rerun.

Contains Chapters 4 and 5 drafts, Chapters 1-3 replacement passages, and an adviser/submission checklist. Editable-manuscript integration and the department template are deferred at the researcher's request. This is not an institution-approved final manuscript.

Reading order

Chapter 4: Results and Discussion

Chapter 5: Summary, Conclusions and Recommendations

Chapters 1-3: Revision package

Adviser review and submission preparation

Chapter 4 Results and Discussion

4.1 Overview of the evaluation

This chapter evaluates whether dictionary-assisted normalization and hybrid retrieval improve responses to code-switched local government service inquiries. It addresses the three research objectives stated in Chapter 1: implementing an approach that captures charter identifiers and conversational intent, comparing the proposed architecture with retrieval baselines, and evaluating answer relevance and factual faithfulness through automated and human assessment. The results support different conclusions at different stages of the pipeline. Hybrid configurations retrieved relevant material more effectively than the raw-query BM25 baseline in the original evaluable subset. The expert audit, however, did not establish statistically significant differences in normalization fidelity or final answer quality among the compared configurations.

The automated evaluation used 70 held-out scenario groups, each expressed in English, Taglish and Bislish. All three architectures processed the resulting 210 queries, producing 630 archived outcomes. A separate development set contained 30 groups and 90 queries. Development data were used for selection of retrieval settings; the held-out outputs were retained without rewriting unsuccessful answers. The human audit covered the fixed category-balanced subset of 20 groups, four each from the Business Bureau, City Treasurer's Office, Office of the City Building Official, Bureau of Fire Protection, and City Planning and Development Office.

Configuration A used the original query with BM25 retrieval. Configuration B used query-specific municipal dictionary injection, constrained normalization and hybrid retrieval. Configuration C disabled query-specific injection while retaining the remaining B pipeline. All configurations used the registered Mistral generator and answer prompt. Configuration C retained shared static lexical guidance, so the B-C comparison isolates the incremental contribution of query-specific injection within that shared guidance rather than every possible form of dictionary knowledge.

The automated results are judgments and retrieval measurements, not proof that all source facts or answers are correct. The human scores are analyzed as independently assigned expert ratings on the researcher's explicit confirmation. Earlier assessment files carried conflicting preparation metadata; the originals, correction record and unchanged numerical scores are retained. The analysis did not independently observe the scoring sessions, and the reliability coefficients do not establish the provenance of collection.

4.2 Implementation of the evaluated architecture

The evaluated corpus contains 1,082 chunks derived from 358 source pages. Preprocessing retained source document and page identifiers and used actual-token windows of 120 tokens with 20-token overlap, with shorter terminal windows where necessary. Table-representation repairs were preserved separately from earlier extraction artifacts. Retrieval operated at chunk level, rather than using section-level repair pilots. These decisions support traceability from a generated response to the passages that were available to the model.

The registered configuration selected K=3 and fusion weights of 0.75 for BM25, 0.25 for dense retrieval and 0 for the dictionary retrieval boost. The zero retrieval-boost weight does not remove dictionary injection before normalization in Configuration B. It indicates that the selected candidate did not assign an additional contribution to the dictionary component of the retrieval score. Generation used temperature 0, top-p 0, an 8,192-token context setting and a maximum prediction setting of 1,536 tokens; normalization used a maximum prediction setting of 150. These settings were recorded for reproducibility, without treating deterministic settings as a guarantee of identical results across every runtime or hardware environment.

The conceptual sequence remains dictionary-assisted normalization, chunk-level retrieval, grounded answer generation and offline evaluation. RAGAS was used after generation; it was not an online judge that repeatedly repaired or regenerated answers. The implementation therefore addresses Objective 1 at the system-construction level. Whether these components improved the measured outcomes is considered separately below.

4.3 Execution coverage and unsuccessful outcomes

Table 4.1 reports all archived generation dispositions. Clarification and incomplete generation are explicit outcomes rather than silently removed examples. The term answered identifies the recorded generation status and does not certify factual accuracy.

Configuration	Planned queries	Answered	Clarification required	Incomplete generation
A	210	209	0	1
B	210	204	5	1
C	210	203	5	2

Table 4.1. Final generation coverage. Source: current completion assessment and archived A/B/C outcomes.

All 630 planned outcomes are present and scoring attempts are complete, but not every metric is observed for every query. A missing judge score is not automatically a zero or a successful judgment. Matched comparisons use the intersection of queries with an observed value for the metric in all three configurations. Consequently, denominators vary across the automated measures and must accompany the reported means.

4.4 Retrieval label versions and automated results

The source-ID crosswalk registered in the execution manifest transferred 77 of 87 legacy source spans and yielded complete retrieval labels for 59 of 70 groups, or 177 queries. A later crosswalk contains ten additional table/source-page reconciliations, extending label coverage to all 210 queries. The later file has a different hash from the crosswalk recorded before inference. Some current summary fields show 210-query coverage while older narrative text still describes 177 queries.

This chapter separates the two label versions. Table 4.2 presents retrieval measurements on the original 177-query complete-label subset and matched RAGAS results on their own observed subsets. The retrieval means were recomputed from per-query values restricted to the original crosswalk's eligible

queries and reproduce the historical complete-case results. Table 4.3 shows the current 210-query reconciled retrieval export separately. This drafting step did not independently recertify the ten later source mappings; their recorded source-page descriptions and hashes remain in the evidence archive. The second table is not presented as if its labels were all registered before inference.

Metric	Matched queries	A	B	C
Hit@3	177	0.5593	0.6667	0.6667
MRR@3	177	0.3588	0.5565	0.5508
Faithfulness	188	0.7481	0.7636	0.7594
Context precision	202	0.8486	0.9191	0.9171
Context recall	202	0.8437	0.8415	0.8448
Answer relevancy	202	0.5794	0.5827	0.5956

Table 4.2. Matched automated results. Hit@3 and MRR@3 use the original complete-label subset. RAGAS denominators are metric-specific shared complete cases. Values are descriptive means.

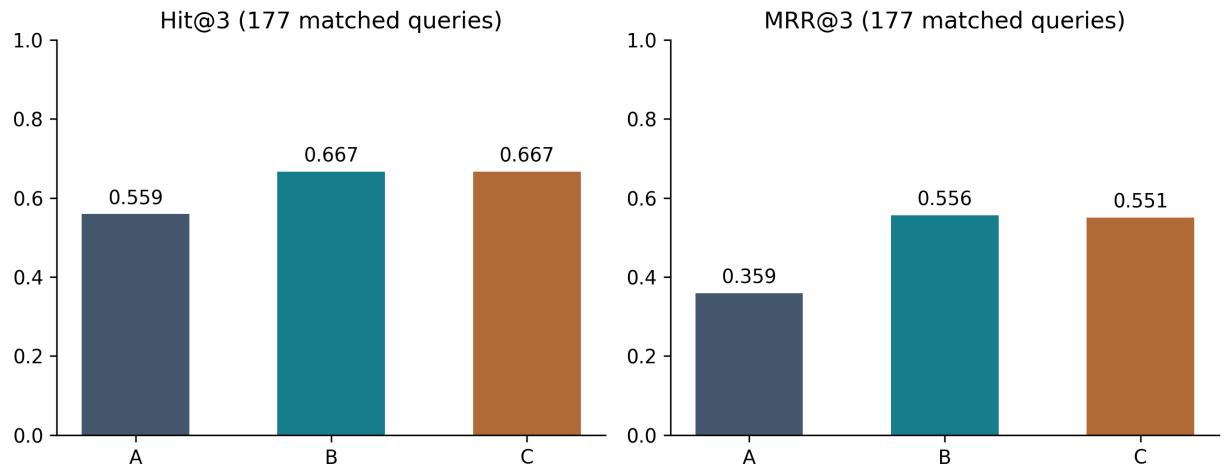


Figure 4.1. Retrieval performance on 177 queries with complete original span-transfer labels. Bar heights are descriptive; no significance test is implied by this figure.

On the original complete-label subset, B increased Hit@3 from 0.5593 to 0.6667 relative to A, a difference of approximately 10.73 percentage points. B also increased MRR@3 from 0.3588 to 0.5565. C had the same Hit@3 as B and a slightly lower MRR@3 of 0.5508. These observations are consistent with a benefit from the normalization-plus-hybrid pipeline over raw-query lexical retrieval, but they do not attribute the entire gain to dictionary injection. B and C share most of the pipeline and their retrieval results are close.

Metric	Matched queries	A	B	C
Hit@3	210	0.5762	0.6810	0.6857
MRR@3	210	0.3659	0.5714	0.5659

Table 4.3. Current retrieval export after the additional ten source-span reconciliations. These 210-query measurements are disclosed as a later label version, separate from Table 4.2.

For automated answer evaluation, B's matched faithfulness mean was 0.7636, compared with 0.7481 for A and 0.7594 for C. B's context precision was 0.9191, compared with 0.8486 and 0.9171. Differences in context recall and answer relevancy did not consistently favor B: C had the largest means for those two measures. The results therefore do not support a blanket statement that B was best on every evaluation criterion.

The local Mistral evaluator has documented semantic and formatting failures. Its role as both generator family and judge family is another limitation. The reported RAGAS means summarize available automated judgments; they should be interpreted alongside missing-score counts, retrieved evidence and the human audit rather than as independently certified factual correctness. No new judge run was used to improve these numbers during manuscript drafting.

4.5 Human audit allocation and normalization results

The adviser-directed amendment introduced two human modules. Module 1 compared the B and C normalized queries for the 20 Taglish and 20 Bislish inputs in the audit subset. Each of three experts scored 80 outputs using the approved five-point statement about preservation of meaning, intent and explicit details, yielding 240 ratings. Module 2 compared the A, B and C final answers for all 60 queries, yielding 180 answer reviews per expert and 540 reviews overall. Two separate four-point dimensions produced 1,080 Module 2 scores. Together the modules contain 780 output-review records and 1,320 dimension scores.

All six supplied assessment files matched the issued packet identities, output IDs, question and output text, presentation order, expected score ranges and corresponding CSV values. No scores were changed or imputed during consolidation. Individual records are retained in a private analysis archive. The different scales were kept separate throughout analysis.

Module / dimension	Config	Scores	Mean	Median
M1 fidelity	B	120	3.7583	4
M1 fidelity	C	120	3.7667	4
M2 relevance	A	180	2.3222	2
M2 relevance	B	180	2.2833	2
M2 relevance	C	180	2.2778	2
M2 faithfulness	A	180	2.7167	3
M2 faithfulness	B	180	2.7278	3
M2 faithfulness	C	180	2.6222	3

Table 4.4. Expert score distributions summarized by configuration. Means use numeric ordinal coding and are accompanied by medians; full score frequencies and language breakdowns are available in the accompanying CSV.

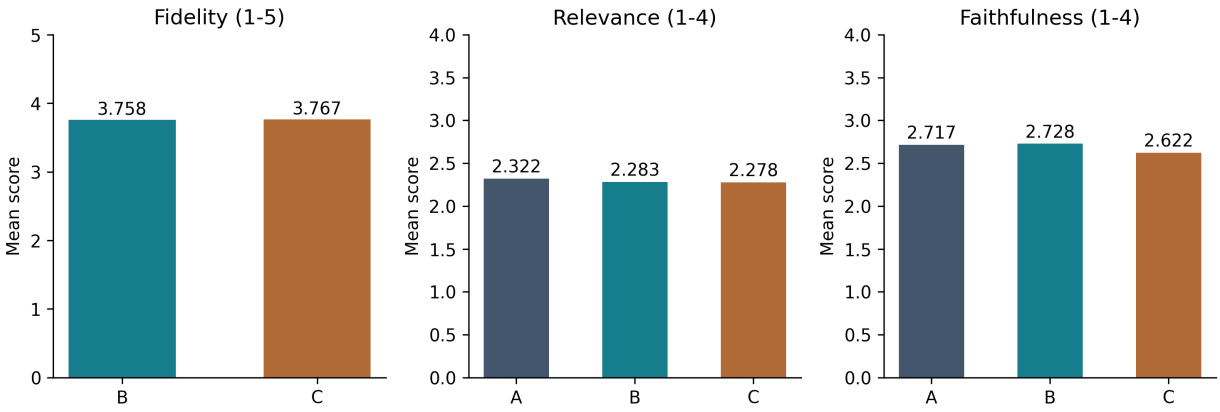


Figure 4.2. Descriptive expert means. Normalization uses a five-point agreement scale; answer relevance and faithfulness use four-point scales. The panels must not be combined into a single cross-module score.

B and C normalization means were 3.7583 and 3.7667, respectively. Both medians were 4. These values indicate that many outputs were judged to preserve the original inquiry, while the distributions still include lower fidelity judgments. The very small aggregate B-C difference does not establish an additional fidelity benefit from query-specific dictionary injection on this sample. Language-specific summaries are descriptive diagnostics and are not additional unadjusted significance claims.

4.6 Answer relevance and factual faithfulness

The answer relevance means were 2.3222 for A, 2.2833 for B and 2.2778 for C, with a median of 2 for all configurations. On Table 2's rubric, a score of 2 describes an answer that addresses part of an inquiry but misses its primary procedural need. Aggregate relevance therefore remains an important practical limitation even when retrieval measurements improve. Access to a relevant chunk does not ensure that a final answer correctly connects the service, requested action, conditions and complete procedure.

Factual faithfulness means were 2.7167, 2.7278 and 2.6222 for A, B and C, respectively; all medians were 3. B had the largest descriptive mean, but the paired tests below did not establish its superiority. The difference between relevance and faithfulness also matters: a response can avoid an unsupported claim while still failing to answer the citizen's actual question. A cautious statement about missing evidence can therefore receive different judgments on the two dimensions.

Seven distinct outputs were flagged as containing no factual claims by at least one reviewer. The primary analysis retains all submitted scores. A separate descriptive sensitivity calculation excludes these outputs and yields faithfulness means of 2.6724 for A, 2.6608 for B and 2.5747 for C. This is not a new primary ranking or a paired test because the denominators differ. It shows that the treatment of no-claim answers can affect descriptive means. The earlier assessment files described an automatic faithfulness convention that is not specified verbatim in Table 2; that history is preserved, and no such automatic rule was applied during this analysis.

4.7 Agreement among expert ratings

Dimension	Outputs	Alpha	95% interval
M1 fidelity	80	0.9354	[0.8945, 0.9641]
M2 relevance	180	0.9112	[0.8590, 0.9424]
M2 faithfulness	180	0.8732	[0.8011, 0.9229]

Table 4.5. Ordinal Krippendorff's alpha. Intervals use 2,000 category-stratified bootstrap draws of scenario groups with all associated languages, outputs and raters retained together.

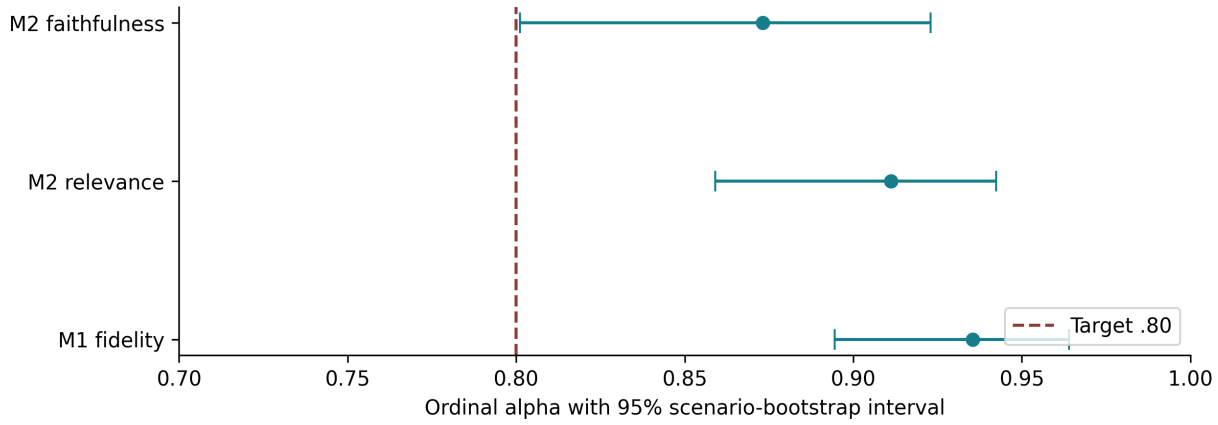


Figure 4.3. Agreement point estimates and scenario-bootstrap percentile intervals. The dashed line marks the manuscript's .80 target, not a significance threshold for system superiority.

The ordinal coefficients were .9354 for normalization fidelity, .9112 for relevance and .8732 for faithfulness. All exceed the manuscript's .80 target. The estimated interval for faithfulness begins near .801, so the lower bound should not be described as comfortably above the target. Bootstrap uncertainty is conditional on the supplied ratings and sampled scenario structure. The observed experts are treated as a fixed panel; the intervals do not represent variation over all possible auditors.

High agreement indicates consistency in the ratings relative to chance disagreement. It does not mean that the systems performed well on every inquiry, that factual references were universally correct, or that collection independence has been independently verified. The implemented ordinal calculation was checked against a separate coincidence-matrix implementation on the actual data and additional test fixtures.

4.8 Paired comparisons

The exploratory comparison plan was recorded before the present computation but after ratings and prior summaries were available. It is therefore not characterized as a preregistered confirmatory analysis. Scores were averaged within each scenario/configuration across the applicable languages and the three raters. This produced 20 paired scenario units, avoiding the mistaken treatment of 780 output reviews as independent scenarios.

Exact two-sided signed-rank tests were used for B-C normalization fidelity and for B-A, C-A and B-C on each answer dimension. Zero differences were omitted from the signed-rank distribution and ties received midranks. A joint Holm correction covered the seven primary comparisons. Group-bootstrap intervals for mean differences are descriptive and not multiplicity-adjusted confidence sets. The test assumes independent scenario groups and a symmetric distribution of differences under the null; a group sign-test sensitivity analysis also showed no p-value below .05.

Dimension	Contrast	Mean difference	95% interval	Raw p	Holm p
M1 fidelity	B-C	-0.0083	[-0.1167, 0.1000]	0.9375	1.0000
M2 relevance	B-A	-0.0389	[-0.2444, 0.1890]	0.3953	1.0000
M2 relevance	C-A	-0.0444	[-0.2444, 0.1778]	0.3085	1.0000
M2 relevance	B-C	0.0056	[-0.0333, 0.0444]	1.0000	1.0000
M2 faithfulness	B-A	0.0111	[-0.3333, 0.3333]	0.8485	1.0000
M2 faithfulness	C-A	-0.0944	[-0.4446, 0.2444]	0.9055	1.0000
M2 faithfulness	B-C	0.1056	[0.0000, 0.2222]	0.2500	1.0000

Table 4.6. Scenario-paired system comparisons. Positive differences favor the first configuration named in each contrast. All tests use 20 scenario groups; nonzero-group counts and rank-biserial effects are supplied in paired_comparisons.csv.

None of the seven comparisons was statistically significant after adjustment. All Holm-adjusted p-values were 1.0000. In particular, B-C faithfulness had only three nonzero scenario differences despite a positive mean difference. This small number of informative pairs limits the strength of a superiority claim. Failure to reject a difference is not evidence of equivalence; the study did not perform an equivalence test with a prespecified practical margin.

4.9 Integrated discussion of the research objectives

Objective 1 was addressed through implementation of source-linked preprocessing, dictionary-assisted normalization, chunk-level lexical/dense retrieval and local grounded generation. Intent-sensitive normalization and explicit handling of ambiguity were part of the implemented workflow. The evaluation nevertheless shows that a working architecture is not sufficient evidence that intent or factual accuracy is preserved for every question.

Objective 2 was addressed by executing A, B and C with common generation settings and preserving their outcomes. Relative to A, the hybrid pipelines showed higher descriptive Hit@3 and MRR@3 on the original complete-label subset. The close B-C retrieval measurements and the absence of a significant human normalization difference mean that the incremental contribution of query-specific dictionary injection remains unestablished in this audit. These findings should be stated without attributing the shared benefits of normalization and hybrid retrieval exclusively to the dictionary.

Objective 3 was addressed through offline RAGAS assessment and expert evaluation against the charter evidence. Expert agreement met the stated target, but relevance means around 2.3 and faithfulness means around 2.6-2.7 leave room for procedural omissions and factual imprecision. The human comparison does not demonstrate superiority of B over A or C. The automated and human results answer related but distinct questions and should not be pooled into a single effectiveness score.

Practical deployment therefore remains a separate decision. The assistant can support retrieval of charter information, but these results do not establish that it can replace official procedural guidance, issue permits, process payments or provide general legal advice. The app defaults were not automatically promoted to the evaluated configuration. Any demonstration should identify the loaded configuration and corpus rather than implying that every local app run reproduces the archived evaluation.

4.10 Limitations of the results

The fixed audit subset was selected after T07 outcomes, although its ordering rule used group identifiers within categories rather than answer quality. Later retrieval-label reconciliation increased coverage and must remain separately identified. The original held-out reference answers were retained and are not universally source-certified. Ambiguous charter text, table extraction and boundary-overlapping source spans can affect what a relevance label represents.

Shared static lexical guidance in C narrows the ablation interpretation. Cross-module answer IDs share opaque suffixes for 80 normalization outputs, allowing linkage, and output similarities can also suggest system identity. Perfect blinding should not be claimed. Earlier conflicting scoring metadata was corrected on the researcher's confirmation; statistical calculations cannot resolve that history independently. No signatures, qualifications or collection timestamps are inferred from the numerical results.

The study uses a small, category-balanced human sample and three observed raters. The inferential analyses are exploratory and do not establish general population effects, equivalence or causal improvement from dictionary injection. No user-adoption, service-time, transaction-success or

legal-compliance outcomes were measured. Those potential benefits remain questions for future evaluation.

Evidence sources for this chapter

The primary methodological source is 20260923.pdf, especially Chapter 1 objectives (PDF pages 7-8), pipeline and evaluation design (PDF pages 28-36), and Table 2. The two-module allocation follows CHAPTER3_AMENDMENT.md dated 26 October 2026. Numerical sources are the final archived matched/per-query metrics, both source-span crosswalk versions, the candidate and execution manifests, and the validated human analysis results. Exact paths and hashes are recorded in source_manifest.json. Statistical implementation references are the Fast Krippendorff project and the SciPy signed-rank documentation, as recorded in the human analysis plan.

Chapter 5 Conclusions Limitations and Recommendations

5.1 Summary of the study

This study implemented and evaluated a dictionary-assisted hybrid retrieval-augmented generation assistant for code-switched local government service inquiries in Davao City. Its workflow combines charter-based preprocessing, municipal terminology assistance, constrained query normalization, chunk-level retrieval and local answer generation. Evaluation compared a raw-query BM25 baseline, the proposed pipeline and an injection-disabled ablation. Automated execution covered 70 held-out scenario groups in three language variants, while the amended expert audit examined a category-balanced subset of 20 groups through separate normalization and answer modules.

The completed implementation and evaluation address the research objectives, but the results do not support an unqualified claim that the proposed system is superior. The hybrid configurations improved descriptive retrieval on the original complete-label subset. Expert ratings showed strong ordinal agreement, yet paired comparisons did not establish significant differences among the tested systems. The distinction between implementation, retrieval success and final answer quality is central to interpreting the study.

5.2 Conclusions by research objective

5.2.1 Capturing charter identifiers and conversational intent

The study demonstrates an implementable architecture for connecting multilingual citizen inquiries to source-linked charter passages without supervised model fine-tuning. Actual-token windows, page provenance and preserved extraction repairs make the evidence used by generation inspectable. Dictionary assistance and constrained normalization provide a mechanism for handling local terminology before lexical and dense retrieval. The registered runtime preserves a reproducible division between preprocessing, retrieval, generation and offline evaluation.

Implementation does not guarantee semantic fidelity. The expert normalization medians were 4 on the five-point scale, but lower-rated outputs remained. Meaning, service, actor direction and explicit qualifiers must therefore be evaluated rather than assumed to survive translation. Ambiguity clarification and retrieval failure should remain distinct: a query may be clear even when the returned evidence is inadequate, and a fluent translation may still change its intent.

5.2.2 Comparing the proposed architecture with baselines

The proposed pipeline had higher descriptive Hit@3 and MRR@3 than raw-query BM25 on the original 177-query complete-label subset. This supports a limited conclusion that the combined normalization and hybrid retrieval workflow found relevant annotated material more effectively in those observations. The later 210-query reconciled label export is reported separately because it extends the original source-label coverage.

The incremental benefit of query-specific dictionary injection was not established. B and C had identical Hit@3 on the original subset and close MRR@3, normalization and answer relevance values. C retained shared static lexical guidance, so this ablation does not represent the total absence of linguistic knowledge. The paired human tests did not demonstrate significant B-C differences. The study consequently cannot claim that dictionary injection alone caused the observed retrieval advantage over A.

5.2.3 Evaluating relevance and factual faithfulness

The study produced both automated and human evidence of answer quality. Ordinal agreement coefficients exceeded .80 for all three human dimensions, indicating consistent ratings on the supplied assessments. At the same time, answer relevance means were approximately 2.3 on the four-point scale and faithfulness means ranged from approximately 2.6 to 2.7. These results identify limitations in procedural completeness and factual precision rather than establishing uniformly reliable service guidance.

No primary human comparison reached statistical significance after the seven-test Holm correction. The appropriate conclusion is that this audit did not establish superior human-rated performance of the proposed configuration. It does not demonstrate equivalence or rule out a difference in a larger, independently planned sample. High rater agreement and high system performance are separate properties and should not be conflated.

5.3 Limitations

The source collection is limited to the studied charter documents and service categories. It does not represent every government service, policy revision or transactional workflow. Source ambiguities and repaired table layouts remain relevant to interpretation. Retrieval-label transfer follows source spans across changed chunk boundaries and can include small boundary overlaps; later reconciliations extend coverage but do not retroactively make all labels part of the original registration.

The generator and offline evaluator used the local Mistral family, and the judge has documented output-format and semantic errors. Missing judge scores were retained and reported. Matched metric denominators differ, so automated means do not represent complete observations for all 210 queries on every measure. The human audit uses only 20 groups and three fixed raters, with exploratory rather than prospectively registered comparison tests.

The audit sample was selected after automated outcomes. The selection rule did not use performance scores, but the timing still limits a claim of preselection. Blinding is also imperfect because content and shared output-ID suffixes can reveal cross-module relationships. The independent-scoring attribution relies on the researcher's explicit confirmation and preserves the earlier conflicting metadata. No statistical coefficient substitutes for collection documentation.

The study did not measure citizen satisfaction, completion time, staff workload, cost savings, legal compliance or actual permit outcomes. Claims that the assistant reduces office congestion, eliminates misleading advice or improves service delivery in practice would exceed the available evidence. Existing app defaults were not automatically synchronized to the evaluated configuration, so operational demonstrations require a configuration check.

5.4 Recommendations

5.4.1 Maintain the municipal dictionary as a versioned resource

The current dictionary should remain a static, versioned component for reproducing the completed study. Each future term revision should retain its source, service scope, governance tags, ambiguity information and review history. Additions should distinguish legitimate alternative senses from incorrect mappings. A new dictionary version should be tested on development data and evaluated under a new declared protocol; the known held-out failures should not be used to silently improve and re-report the frozen test run.

The present results do not justify expanding the dictionary indiscriminately or claiming that every injected definition improves normalization. Future ablations should separate query-specific injection from shared static prompt guidance and measure intent preservation directly. A more discriminating study could preregister service, actor, direction, quantity and condition errors as separate analysis categories while retaining a clear overall fidelity judgment.

5.4.2 Strengthen source representation and evidence coverage

Future source ingestion should continue checking table wording, row relationships, quantities, units, conditions and page references against rendered charters. Uncertain printed values should be preserved and explicitly identified rather than corrected by guesswork. New corpora and indices should be versioned together, with coverage and chunk/index alignment checks retained.

Retrieval and generation errors should be analyzed separately. Wrong-service passages, incomplete context and intent-changing normalization require different interventions. Any new method that carries additional service context must be tested within a prospectively specified development study. It should not be introduced into the already evaluated pipeline and described as the method that produced the archived results.

5.4.3 Plan a stronger follow-up evaluation

A follow-up study should register the audit sample, scoring instructions, no-claim policy, primary comparisons and analysis methods before inspecting final outcomes. Separate module identifiers should prevent direct cross-module output linkage, and collection records should distinguish independent scores from facilitated discussion or later adjudication. Additional scenario groups should be justified by a prospective precision or power analysis rather than selected until a favorable p-value appears.

Evaluator validation should use evidence-linked checks and preserve failed scoring attempts. Agreement should be reported with uncertainty and with clear limitations on generalization beyond the observed raters. If the intended claim is practical equivalence, the study should define an acceptable difference margin in advance and use an appropriate equivalence design rather than interpreting a nonsignificant difference as equality.

5.4.4 Keep deployment claims proportional to evidence

Any operational pilot should identify the exact corpus, dictionary, retriever and model versions in use and provide traceable citations for procedural answers. Staff escalation should remain available when the source is ambiguous or the retrieved evidence is insufficient. A separate authorized pilot would be needed to measure citizen outcomes and staff workload. The prototype should not be represented as an official transaction system or a replacement for authoritative city guidance.

5.5 Final conclusion

The study provides a reproducible implementation and a documented comparison of dictionary-assisted hybrid RAG for multilingual charter inquiries. The hybrid configurations showed descriptive retrieval gains over the lexical baseline on the original evaluable subset, while the expert audit found consistent ratings without statistically established system superiority. The contribution is the implemented workflow, source-linked evidence archive and measured account of its benefits and limitations. Further claims about dictionary-specific benefit, universal factual reliability or real-world service improvement require additional prospective evidence.

Chapters 1-3 revision package

Prepared 28 October 2026 against 20260923.pdf and the Thesis Tracker tab, with the researcher-approved two-module amendment. These are proposed replacement passages for the editable manuscript. The original PDF has not been overwritten. No editable Chapters 1-3 or department LaTeX template was located in the inspected project and usual document locations.

Revision map

Location in original PDF	Required revision	Basis
Chapter 1 objectives, pages 7-8	Preserve the three objectives; distinguish implementation from demonstrated improvement	Actual Chapter 4 findings
Chapter 1 scope, page 8	Replace the 182 integrated barangays category with City Planning and Development Office in the evaluated five-category scope	Issued corpus and scenario allocation
Chapter 1 significance and claims	Express reduced congestion, better service and reduced misinformation as potential benefits, not measured results	No transaction, workload or citizen outcome data
Chapter 2 synthesis	Keep verified original literature and citations; distinguish the proposed dictionary mechanism from empirical proof	B-C differences do not demonstrate superiority
Chapter 3 framework, pages 24-25	Retain normalization, retrieval, grounded generation and offline evaluation	Frozen architecture
Chapter 3 preprocessing, page 28	State actual 120-token windows, 20-token overlap and source/table verification	Evaluated corpus and repair archive
Chapter 3 retrieval/generation, pages 29-31	Report selected K=3 and actual registered weights/settings	Candidate registration and execution manifest
Chapter 3 evaluation, pages 32-34	Separate development selection, held-out execution and later source-label reconciliation	Two crosswalk hashes and archived metrics
Section 3.2.9, page 35	Replace A/B-only allocation with two-module amendment and actual limitations	Adviser mandate and issued triple packets
Analysis, page 36	Describe separate ordinal alpha and exploratory scenario-paired comparisons	Analysis plan and numerical outputs

Chapter 1: proposed scope replacement

The evaluated prototype addresses charter-based service inquiries from five categories: the Business Bureau, City Treasurer's Office, Office of the City Building Official, Bureau of Fire Protection, and City Planning and Development Office. The benchmark contains English, Taglish and Bislish variants of the scenarios. The study evaluates information retrieval and generated guidance from the archived charter corpus. It does not cover every barangay transaction, conduct actual permit processing, or establish legal compliance of an applicant. The system is an informational research prototype rather than an official transaction service.

Chapter 1: proposed contribution and significance wording

This study implements and evaluates a dictionary-assisted hybrid retrieval-augmented generation pipeline for code-switched local government inquiries. Its intended contribution is a reproducible method for preserving query intent, retrieving charter evidence and evaluating the resulting answers. Improvements in office workload, citizen satisfaction and transaction completion are possible future benefits, but they are not measured outcomes of this study. Whether query-specific dictionary injection improves the evaluated outcomes is an empirical question rather than an assumed result.

Retain the existing three research questions and objectives. Do not rewrite them after seeing results to imply that statistical superiority was guaranteed. Replace statements that the prototype eliminates misinformation with statements that it aims to reduce unsupported responses and that remaining errors are assessed.

Chapter 2: targeted editorial changes

Keep the original verified references and theoretical discussion. In the synthesis, distinguish evidence for retrieval augmentation in the literature from evidence for this specific dictionary and corpus. Replace any prediction stated as a demonstrated result with a hypothesis or design rationale. Add a transition explaining that the three-configuration comparison tests the proposed components under shared generation conditions. Do not invent citations, publication details or a new literature review from the experiment results. A source-level bibliography check remains part of final manuscript integration.

Chapter 3: preprocessing and pipeline replacement

The evaluated corpus comprises 1,082 chunks from 358 source pages. Source text and table representations retain document and page provenance. Table repairs are checked against rendered source pages and archived with earlier extractions. Text is divided into actual-token windows of 120 tokens with 20-token overlap; terminal windows may be shorter. Retrieval and fusion operate at chunk level. Source-table repair is distinct from section-based retrieval, which is not part of the evaluated method.

Configuration A retrieves using BM25 from the original query. Configuration B uses query-specific municipal dictionary injection during normalization followed by hybrid retrieval. Configuration C disables query-specific injection while retaining the other B components, including shared static lexical

prompt guidance. This limits the ablation to the incremental query-specific injection mechanism. All configurations use the same registered generator and answer prompt.

Development selection considered the permitted retrieval cutoffs K=3 and K=5. The final registered candidate uses K=3, BM25 weight 0.75, dense weight 0.25 and dictionary retrieval-boost weight 0. Dictionary injection in B remains active despite the zero retrieval-boost weight. The registered Mistral generator uses temperature 0, top-p 0, context setting 8,192 and maximum prediction setting 1,536. Normalization uses temperature 0, top-p 0 and maximum prediction setting 150. RAGAS is an offline evaluator rather than a live response repair loop.

The executed prompt includes citation guidance. The original manuscript's wording concerning two complete format examples is not fully represented in the registered prompt; this implementation discrepancy is disclosed rather than repaired retrospectively in the frozen experiment. Report the runtime, model digest and package versions from the execution manifest and environment records. Do not copy proposed hardware or operating-system specifications as if they were verified execution measurements.

Chapter 3: benchmark and label-version replacement

The development partition contains 30 scenario groups and 90 language-specific queries. The held-out partition contains 70 groups and 210 queries per configuration. All 630 planned outcomes were archived, including clarification and incomplete generation. Missing judge scores remain missing and matched descriptive comparisons identify the observed denominator for each metric.

The source-span crosswalk registered before inference yielded complete retrieval labels for 59 groups and 177 queries. A later reconciliation extended labels to 70 groups and 210 queries. Chapter 4 reports the original complete-label subset as the primary retrieval table and the later reconciled export separately. The crosswalk files and hashes are retained. The later mappings are not represented as if they had all been registered before inference. These label revisions do not constitute a new generation run.

Section 3.2.9: proposed two-module audit replacement

Three auditors evaluate a fixed category-balanced subset of 20 scenario groups, four per category. The subset was selected after automated evaluation using a fixed ordering of group identifiers within categories rather than observed answer quality. This timing is disclosed; the sample was not registered before T07 outcomes. The researcher confirmed that formal scoring had not started when the two-module amendment was approved.

Module 1 evaluates normalized queries from B and C for 40 code-switched inputs: 20 Taglish and 20 Bislish. Each auditor scores both outputs against the statement, "This normalized query preserves the meaning, intent, and explicit details of the original query." The scale is 1 Strongly Disagree, 2 Disagree, 3 Neither Agree nor Disagree, 4 Agree and 5 Strongly Agree. This yields 80 ratings per auditor and 240 overall.

Module 2 evaluates A, B and C responses for all 60 queries, including English. Each auditor reviews 180 responses, yielding 540 response reviews and 1,080 dimension scores across three auditors.

Answer Relevance and Factual Faithfulness retain the separate four-point rubric in Table 2. Together the modules yield 780 output reviews and 1,320 dimension scores. The two scales are never pooled.

Archived questions, outputs and retrieved passages are retained verbatim. System identities are masked and presentation orders randomized. The instrument specifies five sessions: Module 1 followed by four Module 2 blocks of 15 queries, with at least ten-minute breaks between sessions and after the first 20 Module 1 queries. These are protocol requirements; do not claim that actual session timings or break compliance were observed without collection records.

The actual packets contain shared opaque suffixes that permit linkage between 80 cross-module outputs. Content can also reveal similarities. Therefore, the implemented interface does not guarantee perfect blinding. Final exports require scores in the appropriate scale; unfinished drafts may remain blank. Scores are not generated, corrected or imputed during consolidation.

The supplied records are treated as independently assigned scores on the researcher's explicit confirmation. Conflicting preparation metadata in earlier files, the correction record and unchanged scores are preserved. Collection provenance is attributed to this confirmation rather than inferred from statistical agreement. Seven no-claim outputs remain in the primary analysis; a separate descriptive sensitivity analysis reports the effect of excluding them. The earlier automatic scoring convention is disclosed as a limitation rather than silently imposed on the analysis.

Chapter 3: analysis replacement

Ordinal Krippendorff's alpha is calculated separately for Module 1 fidelity and the two Module 2 dimensions. The manuscript's 0.80 target concerns inter-rater agreement, not proof of system superiority. Percentile intervals use 2,000 category-stratified scenario-group bootstrap samples with associated languages, outputs and raters retained together. Inference is conditional on the three observed raters.

The comparison plan is exploratory: it was recorded before the current computation but after ratings and earlier summaries were available. Scores are averaged within each scenario and configuration across the applicable language variants and raters, yielding 20 paired scenario units. Two-sided exact signed-rank tests compare B-C fidelity and B-A, C-A and B-C for each answer dimension. Zero differences are omitted, tied absolute differences use midranks, and Holm adjustment covers all seven primary comparisons. Scenario-bootstrap mean-difference intervals are descriptive and unadjusted for multiplicity. Sign-test sensitivity results are reported separately. Nonsignificance is not interpreted as equivalence.

Integration requirements

These passages must be merged into the editable manuscript using its actual subsection titles and citation style. Recheck Table 2 verbatim, chapter numbering, figure/table numbering, bibliography and cross-references after integration. Preserve the dated original PDF and amendment. T15 remains in progress until the replacement text is applied and checked in the editable source; preparing this package does not complete T16 formatting or T17 approval.

Adviser review and submission preparation

Prepared 28 October 2026. This is a researcher review package, not an approval certificate or a completed submission.

What is ready

• Chapter 4 draft with six result tables, three figures, automated coverage and the complete human analysis.

• Chapter 5 draft stating supported conclusions and recommendations, including a versioned static dictionary.

• Chapters 1-3 replacement passages tied to actual execution and the adviser-directed two-module amendment.

• Aggregate analysis tables, figure sources, evidence hashes and a versioned progress tracker.

• LaTeX chapter fragments and a generic preview wrapper for later integration into the official template.

Key findings to present

Hybrid configurations show higher descriptive retrieval measures than the raw-query BM25 baseline on the original 177-query complete-label subset. Human ordinal agreement is 0.9354 for normalization fidelity, 0.9112 for relevance and 0.8732 for faithfulness. All exceed the stated 0.80 point target. None of seven scenario-paired comparisons is significant after Holm adjustment; all adjusted p-values are 1.0000. The manuscript must not claim demonstrated dictionary-specific superiority, equivalence, or measured service-delivery benefits.

Concrete review decisions

Issue	Proposed manuscript treatment	Review needed
Original 177 versus later 210 retrieval labels	Primary original-subset table; later reconciliation reported separately with hashes	Adviser accepts transparent versioned reporting
Two-module allocation	240 M1 ratings plus 540 M2 reviews; separate five- and four-point scales	Confirm amendment is reflected in institutional research records
Sample timing and imperfect blinding	Disclose post-T07 selection and cross-module linkage	Assess impact on claims and limitations
Rating provenance	Attribute independence to researcher confirmation; preserve metadata correction history	Check actual collection documentation; do not manufacture signatures or dates
No-claim outputs	Retain all submitted scores; separate descriptive sensitivity	Accept limitation of earlier scoring convention
Nonsignificant system comparisons	Report exploratory scenario-paired tests and no established superiority	Confirm conclusions and analysis presentation
CPDO versus barangay scope	Use actual evaluated CPDO category	Confirm Chapter 1 scope correction

T16: deferred by the researcher

The editable Chapters 1-3, bibliography and official department LaTeX template have not been located. On 28 October 2026 the researcher instructed the assistant to skip this integration step and remember it for later. The included fragments are available for future insertion into the official template. Full compilation and page checks remain pending; do not repeatedly ask for these files or mark this step complete. The review PDF is not certified to meet department formatting rules. No final institutional title page, signatures or approval pages have been invented.

T17: adviser and ethics workflow

Send the review package through the project's actual adviser workflow. Record the returned comments, required revisions and approval evidence. Where the institution requires UREB documents or amendments, use its actual forms and authorized process. The current package does not establish ethics approval, exemption, or permission to collect data. No message has been sent to the adviser or employees by this task.

T18: submission checklist

Item	Current disposition
Chapters 4-5 draft and aggregate figures	Prepared
Chapters 1-3 integration into editable source	Pending source files
Department LaTeX compilation and full layout check	Pending template and source integration
Bibliography and citation cross-check	Pending integrated manuscript
Adviser comments resolved and approval documented	External review required
Required ethics/UREB records present	Verify actual institutional requirements and records
Author names, program, title and required submission metadata	Verify against final approved manuscript
Final PDF and required attachments	Assemble after approval
Repository release contents exclude private keys and raw rater records	Check before any publication
Official portal upload and receipt	Not performed

The researcher should retain the final uploaded files, their hashes, confirmation receipt and submission date. Preparation is distinct from submission; no completion should be recorded without those records.